

Here's your inventory sorted into the three primary categories. Components that don't fit these categories (power supplies, cables, development accessories, displays, etc.) are placed in a fourth section for completeness.

1. Microcontrollers / Development Boards (MCU)

- Raspberry Pi 3B
 - Raspberry Pi 4B
 - Raspberry Pi Zero
 - ESP32 Dev
 - ESP32 MC
 - ESP32 CAM
 - ESP8266 NodeMCU
 - Arduino UNO
 - Arduino Nano
 - STM32
 - Shrike Lite
-

2. Sensors

Environmental Sensors

- DHT11
- SN-AHT (Temperature & Humidity)
- KY013 Analog Temperature Sensor
- Pressure Sensor
- Rain Sensor
- LDR Sensor
- Flex Sensor

Gas Sensors

- MQ-2
- MQ-3
- MQ-4
- MQ-5

- MQ-135

Motion & Presence

- PIR Sensor
- Microwave Sensor
- Ultrasonic Sensor

Light & IR

- IR Sensor Module
- Photo IR Sensor
- Laser Receiver Sensor
- Touch Sensor

Position & Motion

- MPU6050
- Hall Effect Sensor

Biometric

- MAX30102

Weight & Force

- Load Cell Amplifier (HX711)

Input Devices

- Potentiometer (POT)
- Small Rotary Potentiometer
- 4×4 Matrix Keypad
- 1-Key Switch Keypad
- Numerical Input Keypad Module
- 24-Key IR Remote Control

Location

- CM GPS Module
 - GPS Antenna
-

3. Actuators

Motors

- MG995 Servo
- MG90S Servo
- SG90 Servo
- BLDC 1000KV Motor
- TT DC Reduction Motor
- N20 Motor
- Stepper Motor
- 5V DC Motor
- RF-500TB Motor
- SP-500187 Gear Motor
- Door Lock Core Motor

Wheels & Motion

- Tyres
- Omni Wheels
- Caster Ball

Relays & Switching

- 1-Channel Relay
- Solid State Relay (SSR)
- SPDT Power Relay

Solenoids & Pumps

- 5/2 Solenoid Valve
- AC Solenoid Lock
- Solenoid Pump

Indicators & Output

- LEDs
- RGB LED
- Buzzer
- LCD 16×2 I2C
- LCD 20×4 I2C
- OLED Display

4. Communication Modules

- NRF Module
- XBee Pro
- XBee USB Adapter Board
- HC-05 Bluetooth Module
- GSM Module (SIM800A)
- GSM Module (SIM900A)
- RF Wireless Transmitter (315 MHz)
- RF Receiver
- RF Transceiver
- ENC28J60 Ethernet Module
- ENC28J0 Controller

5. Motor Drivers

- MX1515 Motor Driver
- TB717A3 Motor Driver
- TB6613FNG Dual Motor Driver
- L293 Motor Driver
- AC Stepper Driver

6. Power Modules

- TP4056 Lithium Battery Charger
 - Li-Ion Battery
 - Orange 18650 Li-Ion Battery Pack
 - Newmowa Li-Ion Battery
 - Battery Holder
 - Buck Converter
 - DC Step-Up Boost Converter
-

7. Programming & Debugging

- ST-Link V2
 - FTDI32
 - USB to UART Converter
-

8. Displays & Camera

- Raspberry Pi Camera
 - OLED Screen
 - LCD 16×2 I2C
 - LCD 20×4 I2C
-

9. Accessories & Development Tools

- SD Card
 - SD Card Reader
 - HDMI to Micro HDMI
 - HDMI to HDMI
 - Micro USB Cable
 - GPIO Extension Board
 - IO Shield
 - I2C Module
 - Breadboard
 - Male Header Pins
 - Female Header Pins
 - USB Adapter
 - RFID Cards
 - Normal Board Switches
 - Push Button (Big)
 - Push Button (Small)
 - Switch Button
-

10. Instruments

- DSO (Digital Storage Oscilloscope)
- P10 Industrial Controller

This organization follows a standard embedded systems lab inventory, making it easier to manage components for workshops, projects, or stock tracking.